

National Programme for Organic Production (NPOP) and Participatory Guarantee Systems (PGS) India also need analysis reports from organic food testing laboratories stating that organic foods contain no residues of pesticides, heavy metals, antibiotics, etc.

The complete compliance with regulations in marking food products as organic is not strictly monitored due to the hand and glove approach between the regulation enforcement agencies and producers. Therefore non-organic foods are often being falsely labelled and marketed as organic foods. There are several chemical and electronic methods in use for the authentication of organic foods, but taking into consideration the easy handling and non-invasive nature of electronic methods, this article only highlights the growth in electronic authentication of organic foods.

Electronic authentication

Currently, there is an increase in consumer awareness about the quality and authenticity of the food. With ever-increasing health problems at present, a growing number of consumers are willing to pay extra for fruits, vegetables, and other food items labelled organic, but whether they're getting what the label claims is another matter. Now scientists studying conventional and organic foods are devising new ways, in general, and electronic methods, in particular, to make sure farms are labelling their products appropriately, which could help prevent organic food frauds and cheating of consumers. Nowadays, for food analysis, electronic sensors are used because they require less time and operate at a lower cost per analysis than other techniques.

1. Authenticating supplier's identity and data on organic food products. Organic food product supplier's digital authentication or e-authentication may be used when referring to the electronic authentication process that confirms or certifies a person's identity and work. In a time, when fraud and identity theft have become rampant, electronic authentication can be a more secure method of verifying the identity of the supplier.

There are various e-authentication methods that can be used to authenticate a supplier's identity as well as data of organic food products ranging from a password to higher levels of security that utilise multi-factor authentication (MFA). When used in conjunction with an electronic signature, it can provide evidence of whether data about organic food products received/ reported has been tampered with after being signed by its original sender.

Under electronic authentication, a quick response code, a type of 2D bar code is used to provide easy access to information with the help of a smartphone or QR scanner. Information like authentication code, grower code, name, location, contact number, Facebook URL, production date, expiry date, and regulatory authority URL about authentic organic growers can be embedded on QR code that can be tracked easily at no cost. The model of adopting a QR code-based labelling mechanism for organic produces drastically better trustworthiness among organic consumers.

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